Computer Organization (CSD206): Assignment 1

Instructions:

- There are 02 questions in this assignment.
- **Assignment submitted after due date and time will not be evaluated and a score of zero will be awarded for this assignment.**
- Upload a **pdf version** of the document.
- Materials copied from the Internet or otherwise will attract penalty as per the course policy.
- You must draw all the diagrams on your own.
- Make proper citations and attach a reference section, if needed.

Due Date: **5:00 pm, November 2, 2018**

Submitting this Assignment:

You will submit (upload) this assignment on Blackboard. Email submissions will not be accepted.

- Write your answer after the given question in this document.
  Name this document as **A1_CO2018_John_Doe.pdf** in case your name is John Doe [You are required to strictly follow the naming convention.]

Grading: This assignment has 6 points (with **weightage of 6%** in your overall 100 points).

Questions

1. Compare and contrast Intel x86 processor and ARM processor based on the following organizational features: (about 2000 words)

a) Data Types
b) Types of Operations
c) Addressing Modes
d) Instruction Formats
e) Instruction sets

2. Write short notes (about 750-800 words each) on the following I/O interfaces:

a) Infiniband
b) Thunderbolt
c) Firewire

Answers

Ans 1:

DATA TYPES

The x86 takes in data types such as 8 (byte), 16 (word), 32 (doubleword), 64 (quadword) and 128 (doublequadword) bits in length. For efficient memory utilisation and increasing flexibility in data structures, there is the flexibility that words do not compulsorily have to be aligned to those addresses which actually happen to be evenly divisible by the number

2. Similarly, the rule for doublewords is that do not compulsorily have to be aligned to those addresses which actually happen to be evenly divisible by the number 4.
Then, the rule for quadwords is that do not compulsorily have to be aligned to those addresses which actually happen to be evenly divisible by the number 8. Following from this, the rule for doublequadwords is that do not compulsorily have to be aligned to those addresses which actually happen to be evenly divisible by the number 16.
But, if the data stored is accessed through a 32 bit bus, transfer of data happens in 32 bit blocks, which must necessarily begin from those addresses which actually happen to be divisible by the number 4. The request for these misaligned values is converted by the processor into a series of requests for bus transfer. The x86 reserves the lowest address for the least significant byte. This is also called as the little-Endian style, and is used in all of the Intel 80x86 machines.
The data types discussed till now were all general data types. Apart from all these, the x86 system also has a wide variety of specific data types that are recognised and operated on by only some specific instructions.
Some examples are Integer, ordinal, unpacked binary coded decimal (BCD), packed binary coded decimal (BCD), near pointer, far pointer, bit field, bit string, byte string, floating point, packed single instruction, multiple data (SIMD).Some of the x86 numerical data types are – byte unsigned integer, word unsigned integer, doubleword unsigned integer, quadword unsigned integer, byte signed integer (twos complement), word signed integer (twos complement), doubleword signed integer (twos complement), quadword signed integer (twos complement), half precision floating point, single precision floating point, double precision floating point, double extended precision floating point.

ARM stands for Advanced RISC Machine. Their processsors give support to types of data which are 8 (byte), 16 (halfword), 32 (word) bits in length. Now, in this case, the normal rule is that all halfword accesses have to be halfword aligned and the word accesses must be word aligned.
The default case for ARM processor is that the address is treated as though it is truncated, and the address bits [1:0] are treated as zero for word accesses and the address bit at [0] is supposed to be treated as zero for access of halfword.
The ARM instructions that load a single word are defined by architecture to rotate right
The transferred data which is word aligned by the address which is non word aligned 1, 2, or 3 bytes on the basis of what the contents of the 2 least significant bits is.
The three data types that we've just mentioned – byte, halfword, and word – can all used for 2's complement signed integers.
Do note that most of the processor implementations of ARM processor do not provide hardware for floating point data types, thereby resulting in saving energy and space. If floating point arithmetic is a necessity in these processors, then it has to be software-implemented. However, the processor does support an optional floating point co-processor, which has the ability to support both double and single precision floating point types of data, as has been defined in IEEE (Institute of Electrical and Electronics Engineers) 754.

TYPES OF OPERATIONS

We get a wide variety of operation types from x86, including, but not limited to, a number of specialised instructions. The aim had always been to provide tools for the writer of the compiler to give him the ability to write machine language conversion of HLL programs, that was optimised.
We have 4 instructions from x86 that support procedure call and return. They are CALL, ENTER, LEAVE and RETURN. We can get knowledge from looking at the support that is provided by these instructions. Upon calling of a new procedure, we perform the following steps:

- Pushing the point of return onto the stack.
- Pushing the frame (current) pointer onto the stack as well.
- Copying the pointer of the stack to be the new value of the pointer of the frame.
- Adjusting the pointer of the stack to allocate a frame.

What the CALL instruction does, is that it pushes the pointer value of the current instruction onto the stack and initiates a jump to the point of entry of the procedure. It does so by putting the address of the point of entry in the pointer to the instruction. The ENTER performs all the steps mentioned above in just a single instruction.

We have a separate set of specialised instructions that actually deal with the aspect of memory segmentation. These are the privileged instructions and may only be executed from the OS.

There are status flags (bits) used in special registers which can consist of certain operations and are applied in conditional branch instructions. Some of the status bits are Carry (C), Parity (P), Auxiliary Carry (C), Zero (Z), Skip (S), Overflow (O), etc.

Intel also introduced the MMX technology, which has an interesting feature – the introduction of saturation arithmetic for 8 and 16 bit operands.

The ARM architecture boasts of a big collection of types of operations. The principle categories are – Load/Store instructions, Branch instructions, Data processing instructions, multiply instructions, parallel addition and subtraction instructions, extend instructions and status register access instructions.

It also has four condition flags (condition codes) which are stored in the program status register. They are Negative (N), Zero (Z), Carry (C) and Overflow (O).

ADDRESSING MODES

In x86, the following addressing modes are present:

- Immediate mode – in this, the operand is also already included within the instruction. The operand may be a byte, word, doubleword, etc.
- Register operand mode – The operand can be on one of the thirty-two bit general registers, one of the sixteen bit general registers, or one of the eight bit general registers.
- Displacement mode – The operand has an offset, which is actually contained inside the instruction set

- Base mode – It specifies that a particular one of the 8, 16 or 32 bit registers contains the EA (effective address).
- Base with displacement mode – It includes a displacement, which is to be added to the base register, which can actually be any of the general purpose registers.
- Scaled index with displacement mode – It includes the displacement that is to be added to any register, and in this case, it is the index register.
- Base with index and displacement mode – it adds the values of the base register, index register and gets the EA by also adding a displacement.
- Base scaled index with displacement mode – It adds the values of the index reg., multiplied by a scaling factor, displacement and the base register.
- Relative addressing – It may be used in control transfer instructions. A displacement is added to (PC), using which we can get the address of the next instruction.

In ARM processors:

- Offest – A value called the offset is added/subtracted from the contents of the Base register, which gives us the memory address.
- Preindex – This memory address is also obtained in the same way as for offset addressing. The memory address is also then written to the Base register. So we can say that the contents of the base register are either decremented or incremented using the offset.
- Postindex – Base register gives the memory address. An offset modifies the value and the result is written back to the base register.

INSTRUCTION FORMATS

The x86 boasts of a wide variety of instruction formats. All other parts of the instruction are variable, but the opcode is always present.

Prefix bytes:

- Instruction prefix – It consists of the LOCK prefix, or it can contain one of the repeat prefixes.
- Segment override – Explicitly states the register to be used by an instruction.
- Operand size – Default operand size is 16/32 bits
- Address size – The memory can be addressed by the processor using either 16 bit or 32 bit addresses.

Instruction fields:

- Opcode – It is one, two or three bytes long. It may also have additional bits to specify if the data is byte or full size, and if a data field must be sign extended that is immediate.
- ModR/M – Addressing information is provided by this byte, and the next. It actually specifies whether a particular operand is in memory or register. If it is found to be in memory, fields within the byte specify the addressing mode to be used. The Mod field is only 2 bits long, and it joins with the R/M field to form 32 possible values: 8 reg, 24 indexing modes.

- SIB – A certain type of encoding of the ModR/M signifies the inclusion of the SIB byte.
- Displacement – If the addressing mode specifier shows that a displacement has been used, an 8, 16, or 32 but signed integer displacement field will be added.
- Immediate – It gives the value of a 8, 16 or 32 bit operand.

All instructions in ARM are 32 bits in length and have a regular format. The first 4 bits are the condition code. The next 3 bits after this specify the type of execution (general). Usually, the next 5 bits constitute an opcode. The leftover twenty bits are for addressing the operand.

- Immediate Constants – The data processing format specifies both an immediate value and a rotate value. This is so that we can get a larger range of immediate values. Here, the 8 bit immediate value is expanded to 32, and right rotated.
- Thumb instruction set – It is a re-encoded subset of the ARM instruction set. It increases the performance of ARM implementations using a 16 bit or narrower memory.
- Thumb-2 Instruction set – The Thumb instruction set created some problems. With its introduction, the user had blend instruction sets by compiling code that was critical to the performance to the ARM and the rest of the thumb overcomes the problems created by the thumb instruction set.
    - In computers that had been optimised for performance, Thumb 2 was 26% smaller than ARM in terms of size. But it was slightly larger than the original thumb.
    - In computers that had been optimised for space, Thumb 2 was 32% smaller than ARM and slightly smaller than the original thumb.

INSTRUCTION SETS

- The x86 is essentially Complex Reduced Instruction Set (CISC) with some limited RISC features. The instruction set is reduced i.e. it has only a few instruction in the instruction set. Many of these instructions are very primitive. More memory requirement. An instruction contains only a single operand or operation.
- It has a hardwired processor. The same program in RISC often requires many more instructions than CISC. Instructions are only LOAD/STORE instructions, and have a fixed length that do not span word boundaries. Instruction execution execution if faster

- The ARM is essentially a Reduced Instruction Set Computer (RISC). The instruction set has a variety of different instructions that can be used for complex operations. Less memory requirement. An instruction contains an entire statement.
- It has a microprogrammed processor. The same program in CISC often requires many lesser instructions than RISC. Instructions are of many types, and have a variable length, that span word boundaries.

Ans 2:

a) Infiniband

It is the Input/Output spec which is directed at the higher spectrum of the server market. It was initially released in '01 (early) and has gotten multiple vendors. To give an eg: IBM zEnterprise mainframe serioes has extensively depended on it for quite a few years now. The standard specifies specifications and an architecture for flow of data between intelligent Input/ Output devices and processors. It has now become a famous interface for the storage area networking and for other such big scale configurations for storage. Essentially, it enables servers, storage which is remote, and other network devices to be attached with a jumble of switches and links. This architecture, which is based on switches, can connect upto sixty four thousand servers, systems of storage and devices of networking.

In spite of the fact that PCI is a solid interconnect technique what's more, keeps on giving expanded rates, up to 4 Gbps, it is a constrained design contrasted with InfiniBand. With InfiniBand, it isn't important to have the fundamental I/O interface equipment inside the server undercarriage. With InfiniBand, remote stockpiling, systems administration, and associations between servers are expert by connecting all gadgets to a focal texture of switches and connections. Evacuating I/O from the server suspension permits more noteworthy server thickness and takes into account a more adaptable and versatile information focus, as independent nodes might be included as required.

In contrast to PCI, which estimates separations from a CPU motherboard in centimeters, InfiniBand's channel configuration empowers I/O gadgets to be set up to 17 meters far from the server utilizing copper, up to 300 m utilizing multimode optical fiber, and up to 10 km with single-mode optical fiber. Transmission rates has high as 30 Gbps can be accomplished.

The channel connectors are insightful gadgets that handle all I/O capacities without the need to intrude on the server's processor. For instance, there is a control convention by which a switch finds all TCAs and HCAs in the texture and allots intelligent delivers to each. This is managed without processor association. The InfiniBand switch incidentally opens up channels between the processor furthermore, gadgets with which it is conveying. The gadgets don't need to share a channel's ability, just like the case with a transport based plan, for example, PCI, which requires that gadgets mediate for access to the processor. Extra gadgets are added to the arrangement by attaching every gadget's TCA to the switch.

Each physical connection between a switch and a joined interface (HCA or TCA) can be bolster up to 16 consistent channels, called virtual paths. One path is held for texture administration and alternate paths for information transport. Information are sent as a flood of bundles, with every parcel containing some bit of the aggregate information to be exchanged, in addition to tending to and control data. In this way, an arrangement of correspondences conventions are utilized to oversee the exchange of information. A virtual path is briefly devoted to the exchange of information starting with one end hub then onto the next over the InfiniBand texture. The InfiniBand switch maps activity from an approaching path to an active path to course the information between the coveted end focuses.

The key elements of Infiniband are – Host channel adapter (HCA), Target channel adapter (TCA), Infiniband switch, Links, Subnet and router.

To represent the way that a few gadgets can send information quicker than another goal gadget can get it, a couple of lines at the two closures of each connection incidentally supports overabundance outbound and inbound information. The lines can be found in the divert connector or in the joined gadget's memory. A different combine of lines is utilized for each virtual path. The host utilizes these lines in the accompanying style. The host puts an exchange, called a work line section (WQE) into either the send or get line of the line combine. The two most vital WQEs are SEND also, RECEIVE. For a SEND task, the WQE indicates a square of information in the gadget's memory space for the equipment to send to the goal. A RECEIVE

WQE determines where the equipment is to put information got from another gadget at the point when that purchaser executes a SEND activity. The channel connector forms each posted WQE in the best possible organized request and creates a consummation line section (CQE) to show the finishing status.

b) Thunderbolt

The latest, and quickest, fringe association innovation to wind up accessible for universally useful utilize is Thunderbolt, created by Intel with joint effort from Apple. One Thunderbolt link can deal with the work already expected of different links.

The innovation consolidates information, video, sound, and power into a solitary rapid association for peripherals, for example, hard drives, RAID (Redundant Array of Independent Circles) exhibits, video-catch boxes, and arrange interfaces. It gives up to 10 Gbps throughput toward every path and up to 10 Watts of capacity to associated peripherals. In spite of the fact that the innovation and its related particulars have balanced out, the presentation of Thunderbolt-prepared gadgets into the commercial center has, as of this composition, just gradually started to create. This is on account of a Thunderbolt-good fringe interface is extensively more unpredictable than that of a basic USB gadget. The original of Thunderbolt items are fundamentally gone for the prosumer (proficient shopper) market, for example, varying media editors who need to have the capacity to move vast volumes of information immediately between capacity gadgets and workstations. As the innovation winds up less expensive, Thunderbolt will discover mass customer utilizes, for example, empowering exceptionally rapid information reinforcements and altering top quality photographs. Jolt is as of now a standard element of Apple's MacBook Pro workstation and iMac PCs.

Not at all like bus based I/O models, each Jolt port on a PC is fit for giving the full information exchange rate of the connection in the two bearings with no sharing of information transmission limit between ports or among upstream and downstream bearings. For correspondence inner to the PC, the Thunderbolt controller incorporates at least one DisplayPort convention connector ports. DisplayPort is a computerized presentation interface standard currently generally embraced for PC screens, workstation shows, also, different illustrations and video interfaces. The controller additionally incorporates a PCI Express change with up to four PCI Express convention connector ports for inner correspondence. The Thunderbolt controller gives access to outside gadgets through one or more Thunderbolt connectors. Every connector

can give a couple of full-duplex channels, with each channel giving up to 10 Gbps toward every path. The equivalent connector can be utilized for electrical or optical links. The electrical link can broaden up to 3 meters, while the optical link can stretch out into the several meters. Clients can interface elite peripherals to their PC over a link, daisy tying in a steady progression, up to an aggregate of 7 gadgets, 1 or 2 of which can be high-goals DisplayPort shows (contingent upon the controller arrangement in the host PC). Since Thunderbolt innovation conveys two full-transfer speed channels, the client can understand high transfer speed on the primary gadget connected as well as on downstream gadgets also.

The Thunderbolt convention physical layer is in charge of connection upkeep counting hot-plug3 identification and information encoding to give exceedingly productive information exchange. The physical layer has been intended to present exceptionally negligible overhead what's more, gives full-duplex 10 Gbps of usable ability to the upper layers. The regular transport layer is the way to the task of Thunderbolt and what makes it appealing as a fast fringe I/O innovation.

The application layer contains I/O conventions that are mapped onto the vehicle layer. At first, Thunderbolt gives full help to PCIe and DisplayPort conventions. This capacity is given by a convention connector, or, in other words proficient embodiment of the mapped convention data into transport layer parcels. Mapped convention bundles between a source gadget and a goal gadget might be steered over a way that may cross various Thunderbolt controllers. At the goal gadget, a convention connector re-makes the mapped convention in a way that is indistinct from what was gotten by the source gadget. The benefit of doing convention mapping along these lines is that Thunderbolt technology– empowered item gadgets show up as PCIe or DisplayPort gadgets to the working arrangement of the have PC, accordingly empowering the utilization of standard drivers that are accessible in many working frameworks today.

c) Firewire

FireWire is a strategy for exchanging data between computerized gadgets, particularly sound and video gear. Otherwise called IEEE 1394, FireWire is quick - the most recent form accomplishes accelerates to 800 Mbps. Eventually, that number is relied upon to bounce to a mind blowing 3.2 Gbps when producers update the current FireWire links.

You can interface up to 63 gadgets to a FireWire transport. Windows working frameworks (98 and later) and Mac OS (8.6 and later) both help it.

Suppose you have your advanced camcorder associated with your home PC. At the point when your PC controls up, it inquiries the majority of the gadgets associated with the transport and allots every one a location, a procedure called Enumeration. FireWire is Plug-and-play, so in the event that you associate another FireWire gadget to your PC, the working framework auto-recognizes it and requests the driver plate. In the event that you've just introduced the gadget, the PC enacts it and begins conversing with it. FireWire gadgets are Hot-pluggable, which implies they can be associated and disengaged whenever, even with the power on.

The first FireWire detail, FireWire 400 (1394a), was quicker than USB when it turned out. FireWire 400 is still being used today and highlights:

- Exchange rates of up to 400 Mbps
- Greatest separation between gadgets of 4.5 meters (link length)

The arrival of USB 2.0 - including exchange accelerates to 480 Mbps and up to 5 meters between gadgets - shut the hole between these contending models. Yet, in 2002, FireWire 800 (1394b) began appearing in customer gadgets, and USB 2.0 was left in the residue. FireWire 800 is prepared to do:

- Exchange rates up to 800 Mbps
- Most extreme separation between gadgets of 100 meters (link length)

The quicker 1394b standard is Backward-compatible with 1394a.

The key contrast among FireWire and USB is that FireWire is expected for gadgets working with significantly more information - things like camcorders, DVD players and advanced sound hardware. FireWire and USB share various attributes however vary in some critical ways. Actualizing FireWire costs somewhat more than USB, which prompted the reception of USB as the standard for associating most peripherals that don't require a fast Bus.

Speed aside, the enormous contrast among FireWire and USB 2.0 is that USB 2.0 is have based, implying that gadgets must associate with a PC with the end goal to impart. FireWire is Peer-To-Peer, implying that two FireWire cameras can converse with one another without experiencing a PC.

FireWire gadgets can be powered or unpowered. FireWire enables gadgets to draw their capacity from their association. Two power conductors in the link can supply control (8 to 30 volts, 1.5 amps most extreme) from the PC to an unpowered gadget. Two Twisted Pair sets convey the information in a FireWire 400 link utilizing a 6-stick design.

FireWire 800 links utilize a 9-stick setup. Six of those pins are the equivalent as the six sticks in the 1394a connector (appeared previously). Two of the additional pins give a "Grounded Shield" to shield alternate wires from impedance, and the third included stick does nothing as of now.

Since FireWire 800 is Backward-Compatible with FireWire 400, there are an assortment of connectors accessible to encourage the blend of the two principles on a similar transport. There are additionally two sorts of FireWire 800 ports accessible: a "Bilingual" port accomodates both FireWire principles, while a B-just port acknowledges just a FireWire 800 connector.

FireWire utilizes 64-bit settled tending to, in view of the IEEE 1212 standard. There are three sections to every parcel of data sent by a gadget over FireWire:

A 10-bit transport ID that is utilized to figure out which FireWire transport the information originated from

- A 6-bit physical ID that recognizes which gadget on the transport sent the information
- A 48-bit stockpiling zone that is fit for tending to 256 terabytes of data for every hub.

FireWire truly sparkles with regards to advanced video applications. Most advanced camcorders or camcorders presently have a FireWire plug. When you join a camcorder to a PC utilizing FireWire, the association is astounding.

An essential component of FireWire is the help of Isochronous gadgets. In Isochronous mode, information streams between the gadget and the host continuously with ensured transfer speed and no blunder remedy. Basically, this implies a gadget like a computerized camcorder can ask for that the host PC sufficiently designate transfer speed for the camcorder to send uncompressed video continuously to the PC. At the point when the PC to-camera FireWire association enters isochronous mode, the camera can send the video in a relentless stream to the PC without anything upsetting the procedure.

You can without much of a stretch alter and make custom video ventures utilizing quick hard drives, an advanced camcorder and a PC. With the correct programming, the PC and the camera discuss, and the PC can download the majority of the video naturally and with immaculate computerized clearness. Since the substance is computerized all the way, there is no loss of value as you take a shot at progressive ages.

References:

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